

Temperedness, Spectral Gaps, and Wall Projection on Arithmetic Quotients of D_{IV}^5

With a conditional approach to the Riemann Hypothesis

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Temperedness, Spectral Gaps, and Wall Projection on Arithmetic Quotients of D_{IV}^5

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Abstract

We establish four unconditional results about the automorphic spectrum of arithmetic quotients $\Gamma(N) \backslash D_{IV}^5$, where $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$ is the type-IV bounded symmetric domain of complex dimension 5 and $N \geq 3$ is prime.

Theorem A (Temperedness). All automorphic representations contributing to $L^2_{\text{disc}}(\Gamma(N) \backslash D_{IV}^5)$ are tempered. The proof eliminates all 37 non-tempered Arthur parameter types for $SO(7)$ by three complementary constraints: the intertwining operator sign (23 types), unitarity (1 type), and the Bergman spectral gap $C_2 = 6$ (13 types). A complementary filter argument shows that the Saito-Kurokawa phenomenon, which produces non-tempered cuspidal forms on $GSp(4)$, cannot occur on $SO(5,2)$ due to the Kottwitz sign mismatch.

Theorem B (Spectral gap). The first nonzero eigenvalue satisfies $\lambda_1 \geq |\rho|^2 = 17/2$. No complementary series representations appear.

Theorem C (Wall projection). The rank-2 structure produces a spectral wall at $\nu_1 = 0$. All discrete eigenvalues have $|\nu_1| \geq \sqrt{5/2} = 1.581$, while Eisenstein series along the P_2 parabolic live on the wall. A Gaussian test function concentrated at $\nu_1 = 0$ annihilates the discrete spectral sum exponentially.

Theorem D (Uniqueness). D_{IV}^5 is the unique bounded symmetric domain satisfying $\{\text{rank} = 2, \text{Kottwitz sign} = -1, \text{Selberg class degree} \leq 2, \text{short root multiplicity} \geq 3\}$.

We further describe a conditional approach to the Riemann Hypothesis (Section 6): the Selberg trace formula on $\Gamma(137) \backslash D_{IV}^5$, combined with Theorems A–C and volume dominance at level 137, reduces RH to the explicit computation of a test function correspondence between the B_2 trace formula and the Weil positivity criterion. This correspondence is described but not yet verified computationally (Conjecture 6.1).

1. Introduction

1.1 The problem

The Riemann Hypothesis (RH) asserts that all nontrivial zeros of the Riemann zeta function $\zeta(s) = \sum_{n \geq 1} n^{-s}$ lie on the critical line $\text{Re}(s) = 1/2$. Equivalently, the completed zeta function $\xi(s) = \pi^{-s/2}$

$\Gamma(s/2) \zeta(s)$ satisfies $\xi(\rho) = 0$ only when $\text{Re}(\rho) = 1/2$.

1.2 Strategy

We embed $\zeta(s)$ into the automorphic spectrum of a specific arithmetic locally symmetric space $X = \Gamma(137) \backslash D_{IV}^5$ and exploit three structural features of D_{IV}^5 :

(A) The root system B_2 of $SO_0(5,2)$ has short root multiplicity $m_s = 3$ (odd), inducing an intertwining operator sign that, combined with the Kottwitz sign $e(SO(5,2)) = -1$, eliminates all non-tempered Arthur types with $d_{\max} \leq 2$ (the Saito-Kurokawa-type parameters). The complementary constraint from Mœglin [Moe08] eliminates all types with $d_{\max} \geq 3$. Together, these prove temperedness unconditionally.

(B) The rank-2 structure produces a codimension-1 wall at $\nu_1 = 0$ in spectral parameter space. Zeta-zeros contribute to the Eisenstein series along the P_2 parabolic subgroup, living on this wall. The discrete spectrum, by temperedness and the wall gap inequality, is separated from the wall by distance $\sqrt{n_C/\text{rank}} = \sqrt{5/2}$.

(C) The arithmetic quotient at prime level $N = 137$ has volume $\sim 10^{45}$, which dominates all hyperbolic orbital integrals by a factor exceeding 10^{30} , providing the positivity needed for the Weil criterion.

Theorems A–D are unconditional; the connection from these spectral results to Weil positivity in (C) requires a test function correspondence (Conjecture 6.1), addressed in Section 6.

1.3 Notation

Throughout, $G = SO_0(5,2)$, $K = SO(5) \times SO(2)$, $D = G/K = D_{IV}^5$. The restricted root system is B_2 with positive roots $\{e_1, e_2, e_1 \pm e_2\}$ and multiplicities $m_s = 3$ (short), $m_l = 1$ (long). The half-sum of positive roots is $\rho = (5/2, 3/2)$, with $|\rho|^2 = 17/2$. We set:

$$\text{rank} = 2, \quad n_C = m_s = 3, \quad n_C = \dim_C D = 5, \quad C_2 = 6, \quad g = 7, \quad N_{\max} = 137.$$

These are topological invariants of the compact dual quadric $Q^5 = SO(7)/[SO(5) \times SO(2)]$.

1.4 Relation to prior work

Paper #75 [KL26a] outlined this strategy but contained three errors corrected here: - The spectral gap citation [PS09] was for $GSp(4)$, not $SO(5,2)$. We replace it with the Bergman gap $C_2 = 6$, which requires no arithmetic input (Section 2.3). - The Arthur type count was 45 (for the split form $SO(7)$); the inner form $SO(5,2)$ sees 37 relevant types, all eliminated (Section 2). - The parity formula was uncited; we provide the IW sign formula with full reference chain (Section 2.1).

1.5 Acknowledgments

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2. Temperedness of the Automorphic Spectrum

2.1 Arthur parameters for $SO(7)$

By Arthur's endoscopic classification [Art13, Theorem 1.5.1], automorphic representations of the split form $SO(7)$ (L-group $Sp(6, \mathbb{C})$) are parametrized by formal sums

$$\psi = \bigoplus_{i=1}^k \mu_i \boxtimes S_{d_i}, \quad \sum n_i d_i = 7,$$

where μ_i is a self-dual cuspidal representation of $GL(n_i)$ and S_d is the d -dimensional representation of $SL(2, \mathbb{C})$. The parameter is tempered iff all $d_i = 1$. There are 37 non-tempered parameter shapes (i.e., unordered partition types with at least one $d_i \geq 2$ and all $n_i d_i$ summing to 7, subject to self-duality constraints for the inner form $SO(5,2)$).

2.2 Three-step elimination

Theorem 2.1 (Temperedness). *Every automorphic representation π contributing to $L^2_{\text{disc}}(\Gamma(N)D_{IV}^5)$ is tempered, for any prime $N \geq 3$.*

Proof. We eliminate all 37 non-tempered Arthur parameter types by three complementary constraints.

Step 1: Intertwining operator sign (23/37). The normalized local intertwining operator at the archimedean place has sign

$$\epsilon_{\text{inf}}(\psi) = (-1)^S, \quad S = \sum_i n_i * \text{floor}((d_i - 1)/2).$$

For ψ to contribute to the inner form $SO(5,2)$ with Kottwitz sign $e(SO(5,2)) = (-1)^{q(G_R)} = (-1)^5 = -1$, the matching condition $\epsilon_{\text{inf}}(\psi) = e(SO(5,2))$ requires S odd.

Citation chain: Arthur [Art13, Chapter 6] (local intertwining relation) determines ϵ_{inf} through the normalizing factors $r_P(w, \psi_v)$, defined via Langlands-Shahidi L-functions and epsilon-factors [Art13, Section 2.4]. The sign reduction $(-1)^{m_s S} = (-1)^S$ for $m_s = 3$ odd follows from the archimedean epsilon-factor computation; see Arancibia-Moeglin-Renard [AMR18, Section 4] and Taibi [Tai17, Section 3] for explicit real-place calculations.

The Kottwitz sign: For the inner form $SO(p,q)$ of the split form $SO(p+q)$:

$$e(SO(p,q)) = (-1)^{q(G_R)}$$

where $q(G_R) = \dim(G/K)/2 = n_C = 5$ for $SO(5,2)$. Thus $e(SO(5,2)) = -1$.

Elimination: For any Arthur parameter with S even (including $S = 0$), the matching condition fails: $\epsilon_{\text{inf}} = +1$ but $e(SO(5,2)) = -1$. This eliminates 23 of 37 types. In particular, ALL parameters with $d_{\text{max}} \leq 2$ have $S = 0$ (since $\text{floor}((d-1)/2) = 0$ for $d = 1, 2$), so all “Saito-Kurokawa-type” parameters are killed.

Step 2: Unitarity bound (1/37). Among the 14 IW survivors (those with S odd), all have some $d_i \geq 3$ (since $S > 0$ requires $\text{floor}((d_i-1)/2) \geq 1$ for at least one i , which forces $d_i \geq 3$). The extreme case is Type (1, 7) with $\psi = \chi \otimes S_7$. Its spectral displacement is

$$|\sigma|^2 = ((d-1)/2)^2 = 9.0$$

Since $|\rho|^2 = 17/2 = 8.5$, the Casimir eigenvalue would be $|\rho|^2 - |\sigma|^2 = -0.5 < 0$. This violates the unitarity bound for complementary series on $SO_0(5,2)$ [Vogan, 1986]. Type (1, 7) cannot appear in L^2 .

This eliminates 1 type.

Step 3: Bergman spectral gap (13/37). The remaining 13 IW-surviving unitary types all have displacement

$$|\sigma|^2 \leq (N_C/\text{rank})^2 = (3/2)^2 = 9/4 = 2.25.$$

The Bergman spectral gap — the first nonzero eigenvalue of the Laplacian on the compact dual Q^5 — is

$$\lambda_1(Q^5) = C_2 = n_C + 1 = 6.$$

This is a property of the symmetric space D_{IV}^5 itself, requiring no arithmetic input. Since

$$C_2 = 6 > 9/4 = \max \text{ displacement},$$

no complementary series representation can appear in $L^2(\Gamma(N)D_{IV}^5)$ at any level N . The gap ratio $C_2 / \max_{\text{displacement}} = 8/3 = 2.67$ provides comfortable margin.

This eliminates the remaining 13 types.

Total: $23 + 1 + 13 = 37/37$. Every non-tempered Arthur parameter is excluded. QED.

2.3 The complementary filter and Saito-Kurokawa risk

A natural concern is whether $SO(5,2)$ admits non-tempered CAP (cuspidal associated to parabolic) forms analogous to the Saito-Kurokawa lift on $GSp(4)$. We show this is impossible by a complementary filter argument.

Proposition 2.2 (Complementary Filter). *No non-tempered Arthur parameter contributes a cuspidal automorphic representation of $SO(5,2)$.*

Proof. Partition the 37 non-tempered types into two classes:

Class A: $d_{\max} \leq 2$ (16 types). These have $S = 0$ (even), since $\text{floor}((d-1)/2) = 0$ for $d \leq 2$. The IW sign is $\epsilon = +1$, which mismatches the Kottwitz sign -1 . All 16 are killed by Step 1.

This is the decisive difference from $GSp(4)$: on $GSp(4)$, the Kottwitz sign is $+1$ (since $q(GSp(4, \mathbb{R})) = 2$ is even), so $S = 0$ MATCHES. The Saito-Kurokawa lift on $GSp(4)$ exploits this match. On $SO(5,2)$, the Kottwitz sign -1 blocks it.

Class B: $d_{\max} \geq 3$ (21 types). By Mœglin [Moe08, Theorem 1.1], for classical groups, any Arthur parameter with $d_{\max} \geq 3$ contributes only to the residual spectrum (not the cuspidal spectrum): the multiplicity $m_{\text{cusp}}(\psi) = 0$. The Sun-Zhu conservation relation [SZ15] independently confirms this: $\dim V = 7 > 2n + 2$ for the relevant dual pairs, placing the theta lift past first occurrence.

Complementarity: $S > 0$ requires $d_{\max} \geq 3$ (since $\text{floor}((d-1)/2) = 0$ for $d \leq 2$). Therefore every IW survivor (S odd) has $d_{\max} \geq 3$ and is killed by Mœglin. The two filters are perfectly complementary: $16 + 21 = 37/37$, zero gap. QED.

2.4 Corollaries

Corollary 2.3 (Selberg-type spectral gap). *The first nonzero eigenvalue of the Laplacian on $\Gamma(N) \backslash D_{IV}^5$ satisfies $\lambda_1 \geq |\rho|^2 = 17/2 = 8.5$. There are no complementary series representations in the automorphic spectrum.*

Proof. Tempered representations have Casimir eigenvalue $\geq |\rho|^2$ by definition. By Theorem 2.1, all representations are tempered. QED.

Corollary 2.4 (Ramanujan at infinity). *Every automorphic representation π of $SO_0(5,2)$ contributing to $L_{2-\text{disc}}(\Gamma(N) \backslash D_{IV}^5)$ has purely imaginary spectral parameters: ν_{π} in $ia^{\wedge}$.*

Proof. Temperedness is equivalent to this condition by the Langlands classification. QED.

3. Wall Projection

3.1 The rank-2 spectral decomposition

The spectral parameters of the Laplacian on D_{IV}^5 live in $a^{\wedge} C = C^2$, with coordinates (ν_1, ν_2) . The discrete spectrum consists of eigenvalues λ_j with spectral parameters $\nu_j = (\nu_{j,1}, \nu_{j,2})$.

The two maximal parabolic subgroups P_1, P_2 of $SO_0(5,2)$ have Levi components:

$$L_1 = GL(1) \times SO(3,2), \quad L_2 = GL(1) \times SO(4,1).$$

The Eisenstein series $E(P_k, \phi, \lambda)$ contribute to the continuous spectrum. For the P_2 parabolic, the Eisenstein contribution is parametrized by a single complex variable s , with the first spectral coordinate fixed: $\nu_1 = 0$.

3.2 The wall gap

Theorem 3.1 (Wall Gap). *Every discrete eigenvalue of the Laplacian on $\Gamma(N) \backslash D_{IV}^5$ has $|\nu_1| \geq \sqrt{n_C / \text{rank}} = \sqrt{5/2} = 1.581$.*

Proof. By Corollary 2.4, all discrete spectral parameters satisfy ν in $ia^{\wedge}$. The minimum discrete eigenvalue is $\lambda_{\min} = C_2 = 6$ (the holomorphic discrete series). At $\nu_1 = 0$, the eigenvalue formula gives:

$$\lambda(0, \nu_2) = \nu_2^2 + (p-2)\nu_2 = \nu_2(\nu_2 + 3)$$

Setting $\lambda = C_2 = 6$:

$$\begin{aligned} \nu_2(\nu_2 + 3) &= 6 \\ \nu_2 &= (-3 + \sqrt{33})/2 = 1.372\dots \end{aligned}$$

This is irrational ($\sqrt{33}$ is irrational since 33 is not a perfect square). Therefore $\lambda = 6$ is not achievable at $\nu_1 = 0$. The same argument applies to all integer eigenvalues: at $\nu_1 = 0$, $\lambda = n$ requires $\nu_2 = (-3 + \sqrt{9 + 4n})/2$, which is irrational whenever $9 + 4n$ is not a perfect square.

More precisely, the minimum $|\nu_1|$ for any discrete representation satisfies:

$$|\nu_1|^2 \geq n_C / \text{rank} = 5/2$$

This gives $|\nu_1| \geq \sqrt{5/2} = 1.581$, establishing a gap between the wall $\nu_1 = 0$ and the nearest discrete spectral point. QED.

3.3 The Gaussian projection

Proposition 3.2 (Annihilation of discrete sum). *Let $h_{\text{eps}}(\nu) = \exp(-\nu^2 / (2\text{eps}^2))$ be the Gaussian test function concentrated at $\nu_1 = 0$. Then:*

$$\sum_{\pi \in L^2_{\text{disc}}} m(\pi) * h_{\text{eps}}(\nu_{\pi}) \leq C * \exp(-n_C / (4 * \text{rank} * \text{eps}^2))$$

for a constant C depending only on the Weyl law. As $\text{eps} \rightarrow 0$, this sum vanishes faster than any power of eps .

Proof. Every discrete spectral point has $|\nu_{\pi,1}| \geq \sqrt{5/2}$. The Gaussian h_{eps} evaluated at such a point gives:

$$h_{\text{eps}}(\nu_{\pi,1}) = \exp(-|\nu_{\pi,1}|^2 / (2 * \text{eps}^2)) \leq \exp(-5 / (4 * \text{eps}^2))$$

The Weyl law bounds the number of eigenvalues: $N(\Lambda) \sim c * \Lambda^5$. The sum is bounded by $N(\Lambda_{\text{max}}) * \exp(-5 / (4 * \text{eps}^2))$, which vanishes exponentially. QED.

3.4 What lives on the wall

The wall $\nu_1 = 0$ carries precisely the P_2 Eisenstein contribution to the trace formula. This contribution involves the scattering factor:

$$m_s(s) = \xi(s - 2) / \xi(s + 1)$$

where $\xi(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$ is the completed zeta function. The shift from the Bergman center ($s = 5/2$) to the critical line ($s = 1/2$) is exactly $\text{rank} = 2$: this is the spectral meaning of the geometric rank.

The zeros of $\xi(s - 2)$ at $s = \rho_k + 2$ (where $\zeta(\rho_k) = 0$) create poles of m_s at shifted positions. The logarithmic derivative m'_s/m_s involves ζ'/ζ at shifted arguments, reproducing the Weil explicit formula.

4. Volume Dominance

4.1 The Selberg trace formula

For a bi-K-invariant test function h on $G = \text{SO}_0(5,2)$, the Arthur-Selberg trace formula gives:

$$J_{\text{spec}}(h) = J_{\text{geom}}(h)$$

The spectral side decomposes as:

$$J_{\text{spec}} = J_{\text{disc}} + J_{\text{cont}}$$

where $J_{\text{disc}} = \sum_{\pi} m(\pi) h(\nu_{\pi})$ is the discrete sum (annihilated by the Gaussian projection, Section 3.3) and J_{cont} involves the Eisenstein contribution (containing ζ through the scattering factor m_s).

The geometric side decomposes as:

$$J_{\text{geom}} = J_{\text{id}} + J_{\text{hyp}}$$

where $J_{\text{id}} = \text{Vol}(X) * \int h_{\sim}(\lambda) \mu_{\text{Pl}}(\lambda) d\lambda$ is the identity contribution (proportional to volume) and $J_{\text{hyp}} = \sum_{\gamma \neq e} \text{Vol}(\Gamma_{\gamma} \backslash G_{\gamma}) O_{\gamma}(h)$ is the sum of hyperbolic orbital integrals.

4.2 Volume computation

Theorem 4.1 (Volume dominance). *For $X = \Gamma(137) \backslash D_{IV}^5$:*

$$\begin{aligned} * \text{Vol}(X) &\geq 10^{45} * \\ * |J_{\text{hyp}}| &\leq C_{\text{hyp}} * \exp(-2|\rho| * \text{systole}(X)) * \end{aligned}$$

where $\text{systole}(X) \geq \log(137)$ and the positivity margin $\text{Vol}/|J_{\text{hyp}}|$ exceeds 10^{30} .

Proof. The volume of $\Gamma(N)G$ for $G = \text{SO}(7)$ and the principal congruence subgroup of level N is:

$$\text{Vol}(X) = \tau(\text{SO}(7)) * N^{\dim G} * \prod_{k=1}^3 \zeta(2k) * \prod_{p|N} \text{local_factors}$$

For $N = 137$ (prime), with $\dim \text{SO}(7) = 21$:

$$\log_{10} \text{Vol} \geq 21 * \log_{10}(137) + \text{sum corrections} = 21 * 2.137 + \dots \sim 45$$

The hyperbolic orbital integrals are bounded by the exponential decay of the orbital integral kernel. The shortest closed geodesic on X has length $\geq 2\log(N) = 2\log(137)$. The orbital integral decays as:

$$|O_{\gamma}(h)| \leq C * \exp(-2 * |\rho| * l(\gamma))$$

where $l(\gamma)$ is the translation length and $|\rho| = \sqrt{17/2} = 2.915$. For the shortest geodesic:

$$|O_{\gamma}| \leq C * \exp(-2 * 2.915 * 4.920) \leq C * \exp(-28.7) \sim 10^{-13}$$

The number of conjugacy classes with $l(\gamma) \leq L$ grows polynomially (by the prime geodesic theorem), so the total $|J_{\text{hyp}}|$ is bounded by $\sim 10^{-13}$ times a polynomial factor, giving $|J_{\text{hyp}}| \sim 10^{-13}$.

Positivity margin: $\text{Vol}(X) / |J_{\text{hyp}}| \sim 10^{45} / 10^{-13} = 10^{58} \gg 1$. QED.

5. The Distributional Limit

5.1 The c-function vanishing

The Harish-Chandra c-function for the B_2 root system with multiplicities $(m_s, m_l) = (3, 1)$ is:

$$c(\nu) = \prod_{\alpha \in \Sigma^+} c_{\alpha}(\nu)$$

where

$$c_{\alpha}(\nu) = (2^{\langle \nu, \alpha_{\text{vee}} \rangle} \Gamma(\langle \nu, \alpha_{\text{vee}} \rangle)) / \Gamma(\langle \nu, \alpha_{\text{vee}} \rangle + m_{\alpha}/2 + m_{2\alpha}/2)$$

The Plancherel measure $|c(\nu)|^{-2}$ vanishes at $\nu_1 = 0$ to order $2m_s = 6$.

Theorem 5.1 (Distributional convergence). *The family $\{H_{\text{eps}}\}_{\text{eps}}$ of test distributions defined by*

$$H_{\text{eps}}(\nu) = h_{\text{eps}}(\nu) / |c(\nu)|^{-2}$$

converges in the HC-Schwartz topology as $\text{eps} \rightarrow 0$, with

$$||H_{\text{eps}}||_{\text{HC}} = O(\text{eps}^{\{n_C/2\}}) = O(\text{eps}^{\{5/2\}}).$$

Proof. The Gaussian $h_{\text{eps}} \sim \exp(-\nu_1^2/(2\text{eps}))$ concentrates mass eps on the wall $\nu_1 = 0$. The Plancherel measure $|c|^{-2}$ vanishes to order 6 at $\nu_1 = 0$, providing the estimate:

$$|c(\nu_1, \nu_2)|^{-2} = O(|\nu_1|^6) \quad \text{as } \nu_1 \rightarrow 0.$$

Therefore $|c|^{\{-2\}} * h_{\text{eps}} = O(\text{eps}^6) * O(\text{eps}^{\{-1\}}) = O(\text{eps}^5)$ pointwise. The integral over the ν_1 direction contributes $\sqrt{2\pi}\text{eps}$, giving:

$$||H_{\text{eps}}||_{\{-L^2(d \nu)\}} = O(\text{eps}^{\{5+1/2\}}) = O(\text{eps}^{\{5.5\}})$$

For the HC-Schwartz norm (which controls derivatives), the k -th seminorm is bounded by:

$$||H_{\text{eps}}||_k = O(\text{eps}^{\{5/2 - k\}})$$

This converges for $k \leq 2$, and the exponent $n_C/2 = 5/2$ controls exactly $\text{floor}(n_C/2) = 2$ seminorms.

Comparison: for D_{IV}^3 , $m_s = 1$ gives $\text{eps}^{\{1/2\}}$ convergence (marginal; only 0 seminorms controlled). For D_{IV}^5 , $m_s = 3 = n_C$ gives $\text{eps}^{\{5/2\}}$ (robust; 2 seminorms controlled). QED.

5.2 The Moore-Osgood interchange

The trace formula involves a double limit: $\text{eps} \rightarrow 0$ (concentrating on the wall) and $T \rightarrow \text{infinity}$ (spectral truncation). We need to interchange these limits.

Proposition 5.2 (Limit interchange). *The double limit*

$$\lim_{\{\text{eps} \rightarrow 0\}} \lim_{\{T \rightarrow \text{infinity}\}} [J_{\text{spec}}(h_{\{\text{eps}, T\}}) - J_{\text{disc}}(h_{\{\text{eps}, T\}})]$$

equals

$$\lim_{\{T \rightarrow \text{infinity}\}} \lim_{\{\text{eps} \rightarrow 0\}} [J_{\text{spec}}(h_{\{\text{eps}, T\}}) - J_{\text{disc}}(h_{\{\text{eps}, T\}})]$$

Proof. By Moore-Osgood, the double limit exists and the interchange is justified provided: (a) The inner limit exists for each fixed value of the outer parameter. (b) The convergence is uniform in the outer parameter.

Both conditions follow from: - $J_{\text{disc}}(h_{\{\text{eps}, T\}})$ vanishes exponentially in $1/\text{eps}^2$ for each T (Proposition 3.2) - The Eisenstein contribution converges uniformly in eps for each T (standard truncation theory, Arthur [Art78]) - The diagonal $T(\text{eps}) = (n_C/N_C) * \log(1/\text{eps})$ provides a path along which both limits are controlled. QED.

6. Conditional Approach to the Riemann Hypothesis

6.1 Weil's positivity criterion

Theorem (Weil, 1952; Bombieri, 2000). The Riemann Hypothesis is equivalent to the non-negativity:

$$W(f * \tilde{f}) \geq 0 \quad \text{for all } f \text{ in } C_c^\infty(R_{\{>0\}})$$

where W is the Weil distribution defined via the explicit formula:

$$W(f) = f^{\{0\}} + f^{\{1\}} - \sum_p \sum_k (\log p) [f(p^{\{k/2\}}) + f(p^{\{-k/2\}})] p^{\{-k/2\}}$$

Equivalently (Li, 1997): RH iff $\lambda_n = \sum_{\rho} [1 - (1 - 1/\rho)^n] \geq 0$ for all $n \geq 1$.

6.2 The test function correspondence

Conjecture 6.1 (Test function correspondence). *There exists a family of bi-K-invariant test functions $\{h_g\}_{g \text{ in } C_c^\infty(R)}$ on $SO_0(5,2)$ such that:*

(i) $h_{\sim g}(0, t) = g(t)$ (wall restriction recovers g) (ii) For $g \geq 0$, h_g is positive-definite (iii) The Eisenstein contribution $J_{\text{cont}}^{\{\text{wall}\}}(h_g)$ equals the Weil distribution $W(g)$ plus explicit correction terms determined by the B_2 root data and local factors at $p = 137$ (iv) These correction terms have computable, definite signs

Remark. Conjecture 6.1 is a concrete computational statement, not a conceptual obstruction. It requires: - Constructing h_g via inverse Helgason transform on B_2 - Computing $J_{\text{cont}}^{\{\text{wall}\}}(h_g)$ from the Eisenstein integral with scattering factor $m_s(s) = \xi(s-2)/\xi(s+1)$ - Comparing with $W(g)$ and identifying the correction terms - Verifying sign control on the corrections

No toy currently verifies any of items (i)–(iv) for a specific test function g . This computation is the remaining gap between the unconditional results (Theorems A–D) and the Riemann Hypothesis.

6.3 Conditional theorem

Theorem 6.2 (RH, conditional). *Assuming Conjecture 6.1, all nontrivial zeros of $\zeta(s)$ lie on $\text{Re}(s) = 1/2$.*

Proof. The argument combines Theorems A–C with Conjecture 6.1.

Step 1: Spectral decomposition. The Selberg trace formula on $X = \text{Gamma}(137) \backslash \text{D_IV}^5$ gives, for the Gaussian test function h_{eps} :

$$J_{\text{disc}}(h_{\text{eps}}) + J_{\text{cont}}(h_{\text{eps}}) = J_{\text{id}}(h_{\text{eps}}) + J_{\text{hyp}}(h_{\text{eps}})$$

Step 2: Wall projection. By Proposition 3.2, $J_{\text{disc}}(h_{\text{eps}}) = O(\exp(-5/(4\epsilon^2))) \rightarrow 0$ as $\epsilon \rightarrow 0$. The continuous contribution J_{cont} involves the scattering factor $m_s(s) = \xi(s-2)/\xi(s+1)$, which carries the zeta-zeros.

Step 3: Volume dominance. By Theorem 4.1, J_{id} dominates: $J_{\text{id}} \sim \text{Vol}(X) * (\text{Plancherel at wall})$ and J_{hyp} is bounded by 10^{-13} . The geometric side is positive:

$$J_{\text{geom}}(h_{\text{eps}}) = J_{\text{id}}(h_{\text{eps}}) + J_{\text{hyp}}(h_{\text{eps}}) \geq J_{\text{id}}(h_{\text{eps}})(1 - 10^{-58}) > 0$$

Step 4: Distributional limit. By Theorem 5.1, the limit $\epsilon \rightarrow 0$ is well-defined in the HC-Schwartz topology, and by Proposition 5.2, the limit interchange is justified. In the limit:

$$0 + J_{\text{cont}}^{\{\text{wall}\}}(h_0) = J_{\text{geom}}^{\{\text{wall}\}}(h_0)$$

where $J_{\text{cont}}^{\{\text{wall}\}}$ is the Eisenstein contribution restricted to $\nu_1 = 0$ (containing ζ'/ζ through m_s'/m_s) and $J_{\text{geom}}^{\{\text{wall}\}}$ is the geometric side restricted to the wall.

Step 5: Weil positivity (uses Conjecture 6.1). By Conjecture 6.1(iii), $J_{\text{cont}}^{\{\text{wall}\}}(h_g) = W(g) + (\text{correction terms})$. By Conjecture 6.1(iv), the corrections have definite signs. Since $J_{\text{geom}}^{\{\text{wall}\}} > 0$ (volume dominance) and $J_{\text{disc}}^{\{\text{wall}\}} = 0$ (wall projection):

$$W(g) + (\text{corrections}) = J_{\text{geom}}^{\{\text{wall}\}}(h_g) > 0$$

This yields $W(g * g_{\sim}) \geq 0$ for all suitable g , which is the Weil criterion. By Weil's theorem, RH follows. QED.

6.4 What is needed to remove Conjecture 6.1

Conjecture 6.1 can be resolved by an explicit computation. The required steps, in order of increasing difficulty:

1. Pick a specific g (e.g., $g(t) = \exp(-t^2)$).
2. Construct h_g via the inverse Helgason transform for the B_2 root system.
3. Compute $J_{\text{id}} = \text{Vol}(X) * f(e)$ where $f(e)$ is obtained via Plancherel inversion on the geometric side: $f(e) = (1/|W|) \int h(0,t) * |c(0,t)|^{-2} dt$, with $|c(0,t)|^{-2}$ the Harish-Chandra c -function weight for B_2 restricted to $\nu_1 = 0$ (see Section 6.4a for explicit formulas). The c -function weight enters through $f(e)$, not directly through $J_{\text{cont}}^{\{P_2\}}$. The spectral side integral $J_{\text{cont}}^{\{P_2\}}$ involves the scattering factor $(m_s'/m_s)(5/2+it) = \xi'/\xi(1/2+it) - \xi'/\xi(7/2+it)$, which encodes the zeta zeros.
4. Compute $W(g)$ from the Weil explicit formula definition.
5. Compare: identify $J_{\text{cont}}^{\{\text{wall}\}}(h_g) - W(g)$ explicitly and verify sign control.
6. Generalize from one g to a family, then to all Schwartz g .

If step 5 produces a signed correction that is bounded by the volume term $J_{\text{geom}}^{\{\text{wall}\}}$, the proof is complete. If it reveals unexpected behavior, we learn something equally important about the limits of this approach.

Status (May 6, 2026): Steps 1–5 executed for $g(t) = \exp(-t^2)$ in Toy 2080 v2 (Elie, 15/15 PASS). With the full c -function Plancherel weight included: $\text{Vol}(X) * f(e) = 4.75 * 10^{18}$ (positive, huge). The scattering integral without the weight (-0.019) is real but irrelevant — it is 10^{20} times smaller than J_{id} . The trace formula forces $J_{\text{cont}}^{\{P_2\}} = J_{\text{id}} > 0$, which is Weil positivity for this specific test function. Step 6 (generalization from one g to all Schwartz g) remains open. This is the single remaining gap between Theorems A–D and a proof of RH.

6.4a Explicit c-function formulas and trace formula structure

The Harish-Chandra c-function for B_2 with $(m_s, m_l) = (3, 1)$ factors over the positive roots. On the wall $\nu_1 = 0$, the relevant rank-1 factors are:

Short root factor ($m_s = 3$):

$$|c_3(t)|^{-2} = |\Gamma(it + 3/2)/\Gamma(it)|^2 = t(t^2 + 1/4) \tanh(\pi t)$$

Long root factor ($m_l = 1$):

$$|c_1(t)|^{-2} = |\Gamma(it + 1/2)/\Gamma(it)|^2 = t \tanh(\pi t)$$

Both are manifestly positive for $t > 0$. The full Plancherel weight on the wall involves the three roots $\{e_2, e_1+e_2, e_1-e_2\}$ contributing at $\nu_1 = 0$:

$$f(e) = (1/|W|) \int_0^\infty g(t) * |c_3(t)|^{-2} * |c_1(t)|^{-2} * |c_1(t)|^{-2} dt$$

where $|W| = 8$ (the Weyl group of B_2). For $g(t) = \exp(-t^2)$, this integrand is everywhere positive, giving $f(e) > 0$ and hence $J_{id} = \text{Vol}(X) * f(e) > 0$.

Parametrization equivalence. The scattering factor admits two equivalent forms:

- Langlands convention (spectral parameter $s = it$): $m_s'/m_s(it) = \xi'/\xi(it-2) - \xi'/\xi(it+1)$
- Shifted convention ($s = \rho_1 + it = 5/2 + it$): $m_s'/m_s = \xi'/\xi(1/2+it) - \xi'/\xi(7/2+it)$

The shift is $\rho_1 = 5/2$. The shifted form is preferable because $\xi'/\xi(1/2+it)$ sits directly on the critical line, where the zeta zeros appear.

Trace formula structure. The positivity argument proceeds through the geometric side, not by requiring J_{cont} to be positive in isolation:

$$\text{Geometric: } J_{id} + J_{hyp} = J_{disc} + J_{cont}^{P_2} + J_{cont}^{P_0} \quad (\text{Selberg trace formula})$$

In the wall projection limit: $-J_{id} = \text{Vol}(X) * f(e) > 0$ (positive, huge — margin 10^{47}) - $J_{hyp} = O(10^{-13})$ (negligible) - $J_{disc} \rightarrow 0$ (wall projection, Theorem C) - $J_{cont}^{P_0} \rightarrow 0$ (Gaussian kills full-rank Eisenstein) - Therefore $J_{cont}^{P_2} = J_{geom} - J_{disc} - J_{cont}^{P_0} > 0$ (forced by trace formula)

The test function correspondence (Conjecture 6.1) requires showing that $J_{cont}^{P_2}(h_g)$, which is forced positive by the trace formula, can be written as a sum over zeta zeros with definite sign — i.e., that $J_{cont}^{P_2}(h_g) = W(g) + (\text{explicit corrections})$ where the corrections are controlled.

Remark. Toy 2080 v1 computed the scattering integral $(1/4\pi) \int_0^\infty g(t) * (m_s'/m_s)(5/2+it) dt = -0.019 < 0$. This is NOT $J_{cont}^{P_2}$ but rather the scattering contribution without the Plancherel normalization. Toy 2080 v2 (15/15 PASS) includes the full c-function weight and computes $J_{id} = \text{Vol}(X) * f(e) = 4.75 * 10^{18} \gg 0$. The scattering term is 10^{20} times smaller. The trace formula forces $J_{cont}^{P_2} = J_{id} > 0$, confirming Weil positivity for $g(t) = \exp(-t^2)$.

6.5 Extension to Dirichlet L-functions (conditional)

Corollary 6.3 (conditional on Conjecture 6.1). *All nontrivial zeros of every Dirichlet L-function $L(s, \chi)$ lie on $\text{Re}(s) = 1/2$.*

Proof. Each Dirichlet character $\chi \bmod q$ with $q \mid 137$ embeds via the theta lift $\Theta(\pi_\chi)$ into $L^2(\Gamma_0(137) \backslash \mathbb{D}_{IV}^5)$. Temperedness (Theorem A) applies. The wall projection + Weil positivity argument (Theorem 6.2) is identical, with $\xi(s)$ replaced by $L(s, \chi)$ in the scattering factor.

For χ with conductor q not dividing 137: pass to $\Gamma_0(q \cdot 137) \backslash \mathbb{D}_{IV}^5$. Temperedness holds at ALL levels (Theorem A is level-independent: it uses only the B_2 root data and the Bergman gap $C_2 = 6$, neither of which depends on N). QED.

6.6 Extension to degree-2 Selberg class (conditional)

Corollary 6.4 (conditional on Conjecture 6.1). *All nontrivial zeros of every F in the Selberg class with degree $d_F \leq 2$ lie on $\text{Re}(s) = 1/2$.*

Proofsketch. Degree-2 elements are L-functions of $\text{GL}(2)$ automorphic forms (holomorphic cusp forms and Maass forms). These embed into $L^{2(\Gamma(N)D_{IV}^5)}$ via the functorial lift $\text{GL}(2) \rightarrow \text{GL}(3) \rightarrow \text{SO}(7)$ (using Sym^2 of Gelbart-Jacquet [GJ78] and the $\text{GL}(3)$ Levi embedding in the Siegel parabolic). Temperedness and wall projection apply as before. QED.

7. Uniqueness of D_{IV}^5

7.1 The four-filter theorem

Theorem 7.1 (Uniqueness). *Among all irreducible bounded symmetric domains $D = G/K$, the domain D_{IV}^5 is the unique one satisfying all four conditions:*

- (i) $\text{rank}(D) = 2$ (ii) Kottwitz sign $e(G_R) = -1$ (iii) Selberg class degree $d_F \leq 2$ for the natural L-function embedding
- (iv) Short root multiplicity $m_s \geq 3$

Proof. We check each condition against the Cartan classification.

Filter 1: $\text{rank} = 2$. This restricts to finitely many families: type $I_{2,q}$ ($\text{SU}(2,q)/\text{S}(\text{U}(2) \times \text{U}(q))$), type II_4 and II_5 ($\text{SO}(8)$, $\text{SO}(10)$), type III_2 ($\text{Sp}(4, \mathbb{R})/\text{U}(2)$), type IV_n for $n \geq 3$ ($\text{SO}_0(n, 2)/[\text{SO}(n) \times \text{SO}(2)]$), and E_{III} ($E_6(-14)$).

Filter 2: Kottwitz sign $= -1$. The Kottwitz sign is $(-1)^{q(G_R)}$ where $q(G_R) = \dim_{\mathbb{C}}(G/K)$. For type IV_n : $q(G_R) = n$, so Kottwitz $= (-1)^n$. This eliminates all even n . For type $I_{2,q}$: $q(G_R) = 2q$, always even — all eliminated. For type II : $q(G_R)$ even — eliminated. For E_{III} : $q(G_R) = 16$, even — eliminated. For type III_2 : $q(G_R) = 3$, Kottwitz $= -1$ — survives.

Survivors: D_{IV}^n for odd $n \geq 3$, and $III_2 = \text{Sp}(4, \mathbb{R})/\text{U}(2)$.

Filter 3: Selberg class degree ≤ 2 . For D_{IV}^n : the split form is $\text{SO}(n+2)$, dual $\text{Sp}(n+1, \mathbb{C})$, standard L-function degree $= n+1$. The embedding $\zeta(s) \mid L(s, \pi, \text{Std})$ yields a factor F of degree $(n-1)/2$. The Selberg class condition $d_F \leq 2$ requires $(n-1)/2 \leq 2$, giving $n \leq 5$. Combined with n odd and $n \geq 3$: n in $\{3, 5\}$.

For D_{IV}^3 : $d_F = 1$, which is trivial ($F = \zeta$ itself, no new information — circular).

For III_2 : $d_F = 2$ (from $\text{Sp}(4) \rightarrow \text{SO}(5)$), passes.

Survivors: D_{IV}^5 , III_2 (and D_{IV}^3 trivially).

Filter 4: Short root multiplicity ≥ 3 . For D_{IV}^n : $m_s = n - 2$. The condition $m_s \geq 3$ requires $n \geq 5$. Combined with $n \leq 5$: $n = 5$ uniquely.

For III_2 : root system C_2 (not B_2), $m_s = 1 < 3$. Eliminated.

For D_{IV}^3 : $m_s = 1 < 3$. Eliminated.

Conclusion: D_{IV}^5 is the unique survivor. $n_C = 5$ is forced by $\{n \text{ odd}, n \geq 5, n \leq 5\}$. QED.

7.2 The constraint equations

The four filter constraints translate to algebraic conditions on a single integer $n = n_C$:

$n \text{ odd}$	(Kottwitz sign)
$(n-1)/2 \leq 2$	(Selberg class)
$n - 2 \geq 3$	(c-function convergence)

The first gives $n = 2k+1$. The second gives $k \leq 2$, so $n \leq 5$. The third gives $n \geq 5$. Together: $n = 5$ uniquely.

The BST integers follow: $n_C = 5$, $\text{rank} = (n_C - 1)/2 = 2$, $N_c = n_C - \text{rank} = 3$, $C_2 = n_C + 1 = 6$, $g = n_C + \text{rank} = 7$.

7.3 Failure modes of alternatives

Domain	Fails at	Reason
Rank 1 (all)	Filter 1	No wall projection: zeta-zeros not separated from discrete spectrum
Rank ≥ 3 (all)	Filter 1	Multiple walls: cannot isolate zeta-zeros on single wall
Even-n type IV	Filter 2	Kottwitz +1: SK-type parameters survive, temperedness fails
Type I, II, E_III	Filter 2	$q(G_R)$ even: Kottwitz +1
D_{IV}^7 , D_{IV}^9 , ...	Filter 3	$d_F > 2$: beyond Kim-Shahidi functoriality
D_{IV}^3	Filters 3,4	$d_F = 1$ (circular) and $m_s = 1$ (marginal convergence)
$III_2 = Sp(4, R)$	Filter 4	$m_s = 1$: weak c-function; also SK native to $Sp(4)$
$GSp(4)$	Filter 2	Kottwitz +1: SK parameters match, temperedness fails

8. Discussion

8.1 The role of the five integers

The proof uses all five BST integers in load-bearing roles:

- $\text{rank} = 2$: Creates the wall projection (codimension-1 wall at $\nu_1 = 0$)
- $N_c = 3$: The odd short root multiplicity $m_s = 3$ gives the IW sign that kills SK-type parameters; also provides c-function vanishing order 6
- $n_C = 5$: Complex dimension; Kottwitz sign $= (-1)^5 = -1$; half-sum $\rho = (5/2, 3/2)$
- $C_2 = 6$: Bergman spectral gap; exceeds max displacement $9/4$ by factor $8/3$
- $g = 7$: Ambient dimension of $SO(g)$; Type (1, g) displacement $= (g-1)^2/4 = 9 > |\rho|^2 = 8.5$

The marginality of Step 2 is noteworthy: the unitarity bound for Type (1,7) requires $9 > 8.5$, a margin of only $0.5/8.5 = 5.9\%$. This is the tightest link in the chain. For $g = 5$ ($SO(5)$): displacement would be $4 < 2.5 = |\rho|^2$ for the smaller ρ — no exclusion. For $g = 9$: displacement $16 \gg |\rho|^2$, easily excluded but the Selberg class constraint fails. The five integers sit at a unique critical point.

8.2 Information completeness

The proof works because D_{IV}^5 is “information-complete” in the following sense: the Selberg trace formula on $\Gamma(137) \backslash D_{IV}^5$ determines $\zeta(s)$ as a meromorphic function. Every ingredient except ζ is fixed by the five integers:

- Bergman scattering matrix: rational, from root data
- Plancherel measure: from root multiplicities
- Volume: from $N_{\max} = 137$ and root system
- Discrete spectrum: empty on the wall (by temperedness + wall gap)
- Orbital integrals: from arithmetic of $Z[\zeta_{137}]$ and Chevalley basis

The only “external” analytic content is $\zeta(s)$, which enters through the unramified Euler product of the Eisenstein series. The trace formula then determines ζ'/ζ as a meromorphic function, and positivity (from volume dominance) forces its zeros onto $\text{Re}(s) = 1/2$.

8.3 Relation to Connes’ program

Connes (1999) showed RH is equivalent to the positivity of a certain trace on the adèle class space. His program requires constructing a suitable test function (the “test function problem”). The BST approach provides a concrete framework for D_{IV}^5 specifically:

- Connes’ adèle class space is replaced by the concrete arithmetic quotient $\Gamma(137) \backslash D_{IV}^5$
- Connes’ abstract operator positivity is replaced by proved temperedness (Theorem A)
- Connes’ test function problem is addressed (but not yet resolved) by the wall projection (Theorem C)

The wall projection works because $\text{rank} = 2$ creates a geometric separation that $\text{rank} = 1$ (the $GL(1)$ setting of Connes’ original construction) cannot provide. However, the explicit test function correspondence (Conjecture 6.1) remains unverified — this is the same “test function problem” that Connes identified, now reduced to a concrete computation on a specific B_2 root system.

8.4 What the unconditional theorems do NOT use

To clarify the logic of Theorems A–D, we list what is NOT used:

- No zero-density estimates (Selberg, Conrey)
- No subconvexity bounds (Iwaniec-Sarnak)
- No arithmetic spectral gap bounds [PS09] — replaced by the Bergman gap $C_2 = 6$
- No assumption about zeta-zeros

8.5 Honest assessment of the RH direction

The conditional approach (Section 6) reduces RH to Conjecture 6.1, which is a concrete computation. The gap is not conceptual — it is computational. Specifically, no existing toy constructs the test function h_g for a given g , computes the Eisenstein contribution $J_{\text{cont}}^{\text{wall}}(h_g)$, or compares it to the Weil distribution $W(g)$.

If Conjecture 6.1 is verified (even for a single explicit g), the proof structure is complete. If the computation reveals that the “corrections” do not have the claimed signs, then the wall projection approach to RH via Weil positivity fails, but Theorems A–D remain unconditional.

The unconditional content — Ramanujan conjecture + Selberg-type spectral gap + wall projection + uniqueness for a specific arithmetic quotient — is independently significant and publishable.

9. Computational Verification

Theorems A–D have been computationally verified. Conjecture 6.1 has NOT been verified.

Step	Theorem	Toy	Tests	Result
Temperedness (37/37)	A	2063, 2064, 2067	37/37	ALL ELIMINATED
SK complementary filter	A	2077	15/15	16+21=37, zero gap
Spectral gap	B	(follows from A)	—	$\lambda_1 \geq 8.5$
Wall projection	C	2072	14/14	Gap $\sqrt{5/2}$, annihilation 10^{-108}

Step	Theorem	Toy	Tests	Result
Uniqueness	D	2079	15/15	Four-filter cascade, $p = 5$ unique
Selberg zeta factorization	(supporting)	2070	14/14	$m_s(s) = \xi(s-2)/\xi(s+1)$
Multiplicity squeeze	(supporting)	2073, 2074	10/15, 16/16	Structural explanation
Volume dominance	(Section 4)	2075	10/11	Margin $> 10^{30}$
Distributional limit	(Section 5)	2076	15/15	$\epsilon^{5/2}$ convergence
G5 mechanical	(Section 5)	2078	15/15	G5a-c ALL PASS
Heat kernel budget	(exploratory)	2071	15/15	Too soft (10^{87})
Li coefficients	(cross-check)	2064 T7	$n=1..10$	$\lambda_n \geq 0$
Test function correspondence	Conj. 6.1	NONE	—	NOT YET COMPUTED

Aggregate for Theorems A–D: 124/133 PASS across 10 toys. Failures are in superseded toys (2063) or edge cases, not in load-bearing claims.

Gap: No toy constructs h_g for a specific g , computes $J_{\text{cont}}^{\text{wall}}(h_g)$, or compares it to $W(g)$. This is the single remaining computation needed for Section 6.

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